

Substitution and Elimination Reactions (S_N2/S_N1, E2/E1)

Use the following two tables as a general rule of thumb

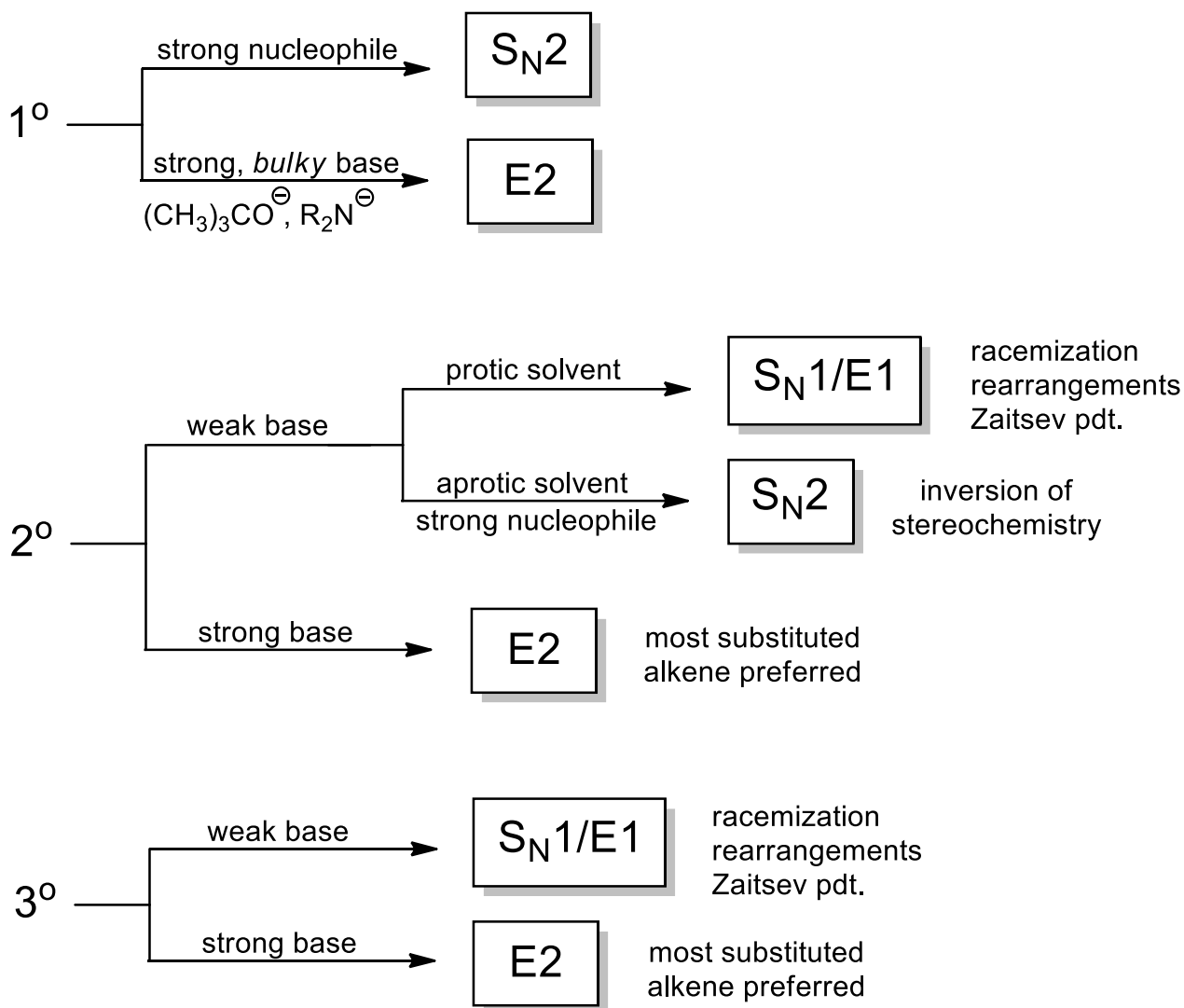
Nucleophilic Substitutions

	S _N 1	S _N 2
Characteristics		
Rate law	k[R-X]	k[R-X][Nuc]
Stereochemistry	racemization	inversion
Rearrangements	common	never
Promoting Factors		
Nucleophile	weak is OK and common	strong
Substrate	3° > 2°	CH ₃ X > 1° > 2°
Solvent	protic	polar aprotic

Eliminations

	E1	E2
Characteristics		
Rate law	k[R-X]	k[R-X][Nuc]
Alkene product	most substituted	most substituted
Stereochemistry	no geometry req'd	coplanar transition state
Rearrangements	common	never
Promoting Factors		
Base	weak is OK and common	strong
Substrate	3° > 2°	3° > 2° > 1°
Solvent	protic	polar aprotic

The following page provides a flow chart used to decide what type of reaction occurs based upon reaction conditions.



The flow chart above is designed to bring it all together for you, conceptually. The first step is to designate the degree of substitution of the substrate then assess the strength of the base and if needed the protic or aprotic nature of the solvent.

**The strength of the base usually determines the order of the reaction. If a strong base is present, second order kinetics follow (S_N2 or E2). If no strong base or strong nucleophile are present, then a mixture of S_N1/E1 occurs.*

***Some nucleophiles and bases favor S_N or E chemistry. To promote elimination, often times bulky bases (like (CH₃)₃CO⁻) are used and can result in formation of the least substituted alkene. To promote substitution with 2° substrates, good nucleophiles which are weak bases are used (RS⁻, NC⁻, I⁻, Br⁻, N₃⁻).*